

# API SECURITY RISK ANALYSIS REPORT

## TARGET WEBSITE

<https://jsonplaceholder.typicode.com/users>

Cyber Security Internship – Future Interns  
**Task 3**

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# OBJECTIVE

The objective of this task is to analyze a public API for potential security risks, identify areas of data exposure, assess authentication controls, and provide recommendations to improve API security.

# TOOLS USED

- Postman
- Browser Developer Tools
- Public Demo API
- Canva

# INTRODUCTION TO API SECURITY

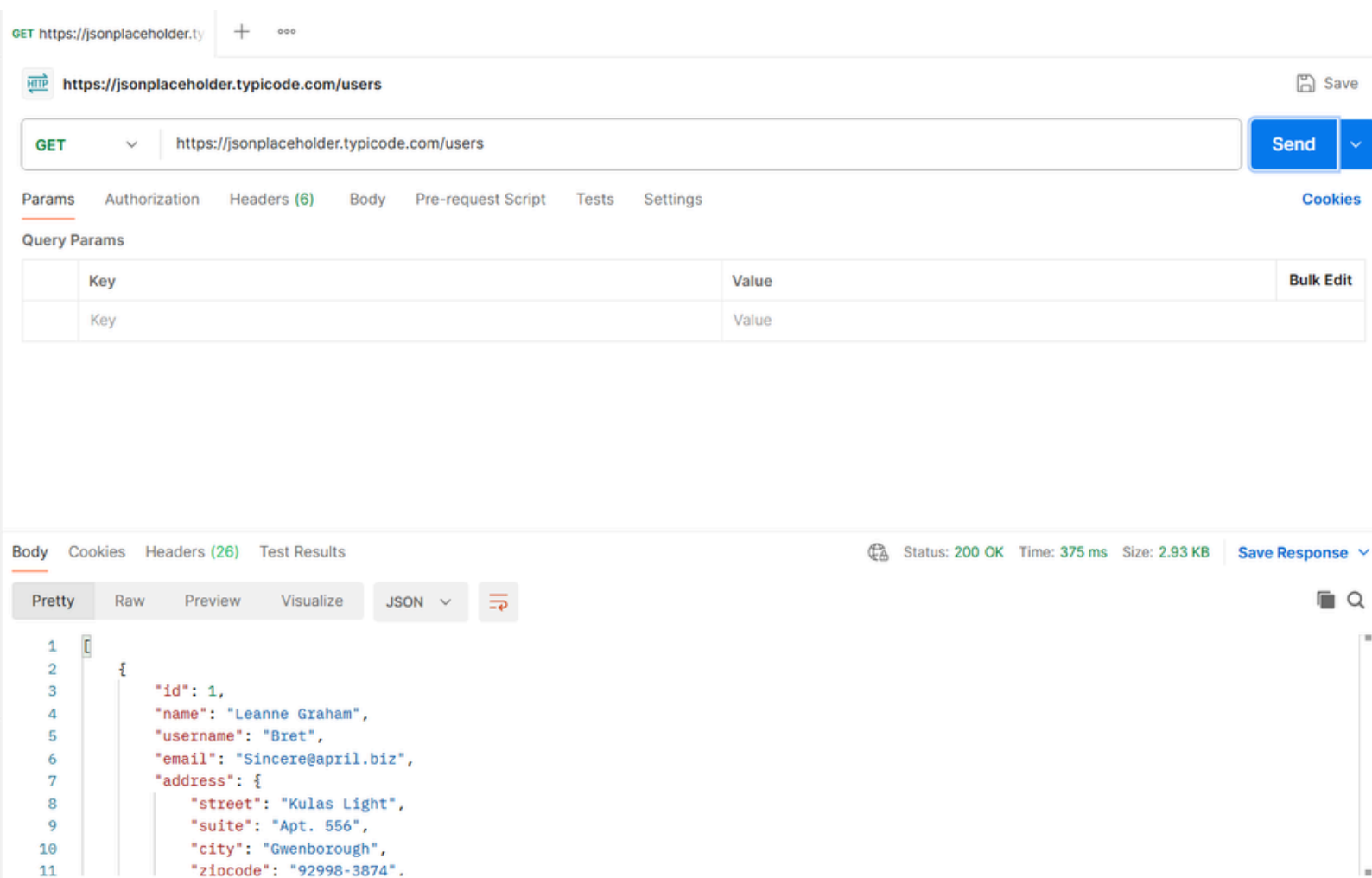
An Application Programming Interface (API) enables communication between different software applications by allowing them to exchange data and services. APIs are widely used in web applications, mobile apps, and cloud-based platforms.

API security is important because APIs often handle sensitive information and business operations. Poorly secured APIs may expose data, allow unauthorized access, or increase the risk of cyberattacks. Regular API security assessments help organizations identify potential vulnerabilities and improve the protection of their systems and data.

# METHODOLOGY

1. Accessed a public demo API using Postman.
2. Sent GET requests to retrieve available API data.
3. Examined the API response for exposed user information.
4. Assessed authentication requirements through the Authorization settings.
5. Analyzed potential security risks related to data exposure and access control.
6. Documented observations and provided security recommendations based on the findings.

# FINDING 1: PUBLIC DATA EXPOSURE



**Figure 1:** API response displaying publicly accessible user information.

## Finding 1: Public Data Exposure

User details such as names, emails, and phone numbers were publicly accessible through the API.

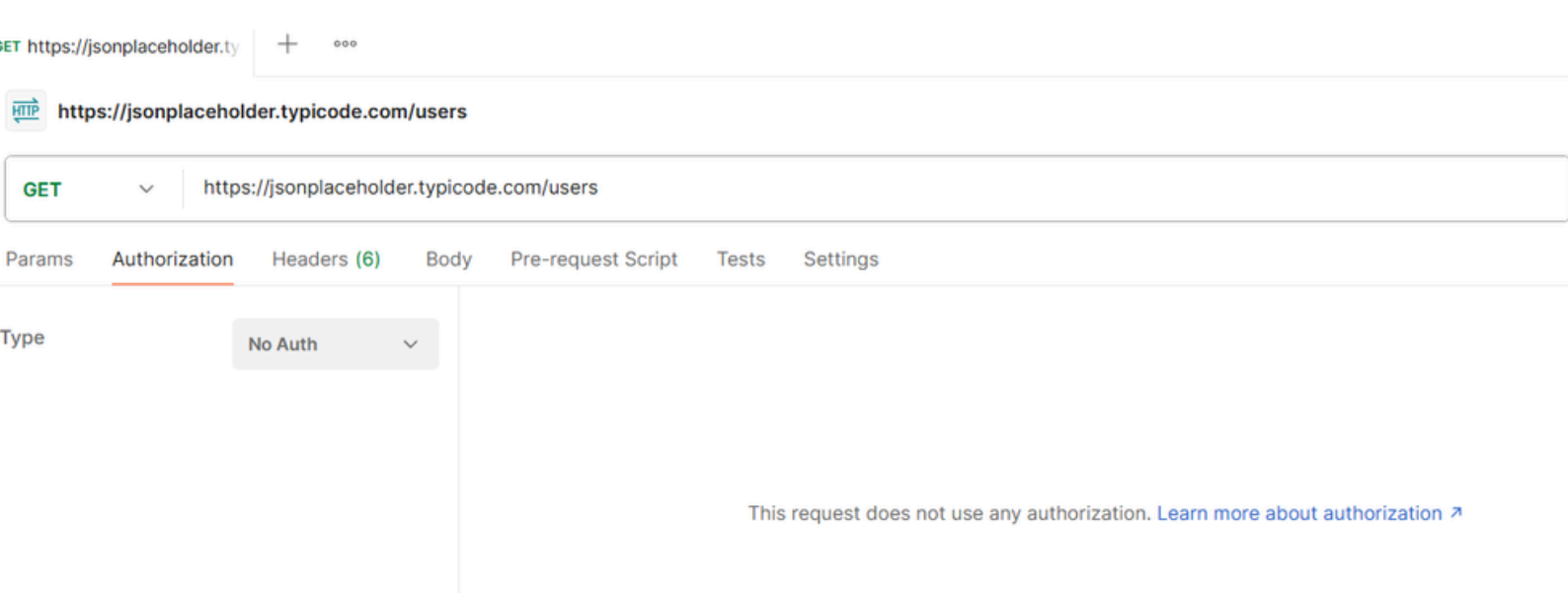
## Risk Impact

Data may be viewed by unauthorized users.

## Recommendation

Restrict unnecessary data exposure and enforce access controls.

# FINDING 2: MISSING AUTHENTICATION



**Figure 2:** Authorization settings showing that no authentication was required.

## **Finding 2:** Missing Authentication

The API endpoint was accessible without any authentication mechanism.

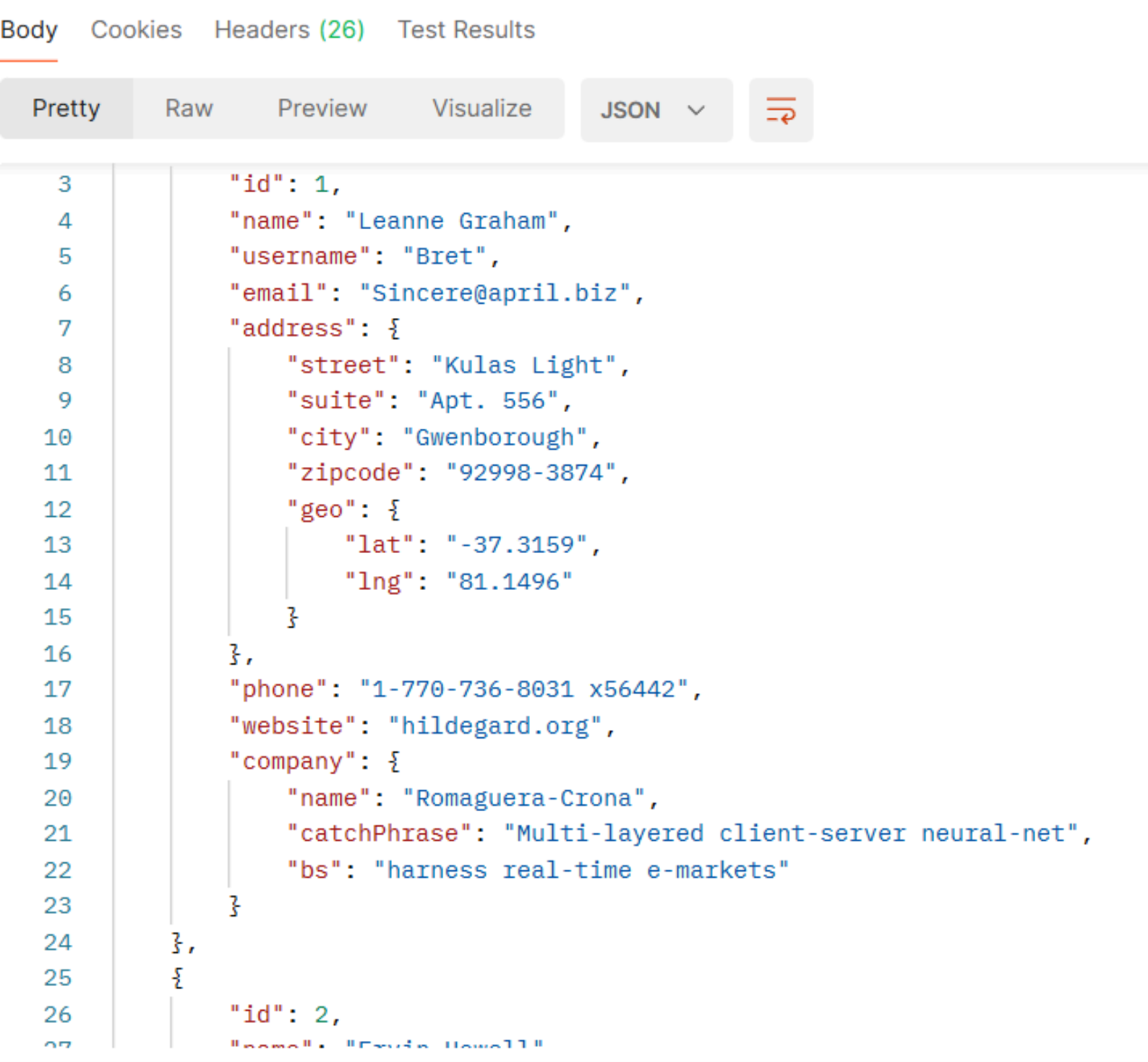
### **Risk Impact**

Unauthorized users may access the available data.

### **Recommendation**

Implement secure authentication methods such as API keys or tokens.

# FINDING 3: MULTIPLE USER RECORDS EXPOSURE



**Figure 3:** API response displaying multiple user records in a single request.

### **Finding 3: Multiple User Records Exposure**

The API returned multiple user records through a single request.

#### **Risk Impact**

Large amounts of information can be collected at once.

#### **Recommendation**

Apply access controls and limit unnecessary data disclosure.

# RECOMENDATIONS

- Implement authentication mechanisms such as API keys or tokens.
- Restrict access to sensitive or unnecessary information.
- Apply proper access control policies for API endpoints.
- Regularly review API responses to prevent excessive data exposure.
- Perform periodic security assessments and monitoring.

# CONCLUSION

This API Security Risk Analysis identified key security concerns, including public data exposure, missing authentication, and the availability of multiple user records through a single request. The assessment highlights the importance of implementing proper authentication, access controls, and data protection measures to reduce security risks and improve API security.

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